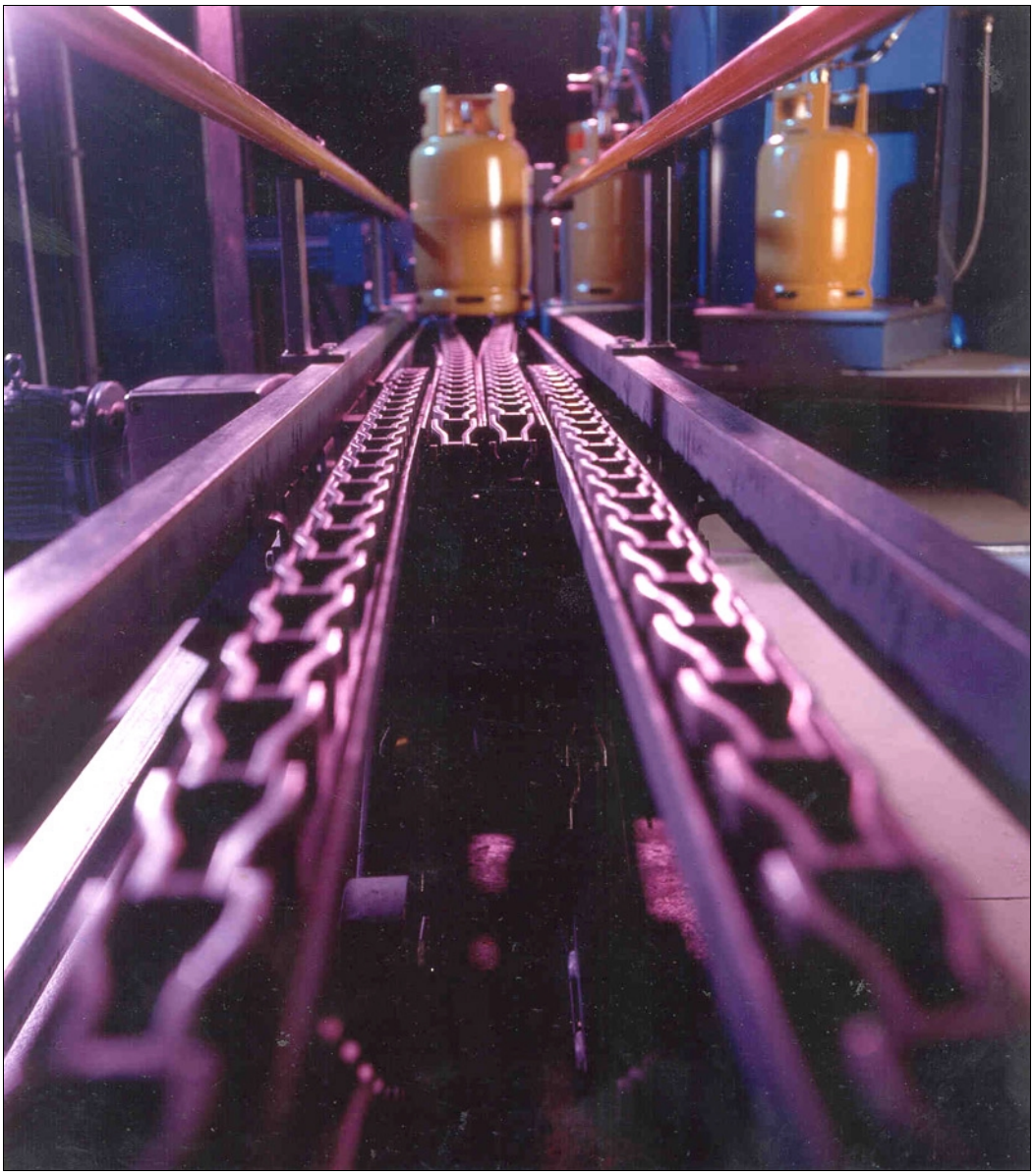


© Kosan Crisplant 2003

General description

The chain conveyor plant is designed for efficient and rational transport of LPG cylinders from the unloading point, passing various points of operation, to the loading point.

A chain conveyor plant consists of the following main components:

- Chain conveyor sections (galvanized) and roller conveyors
- Galvanized driving unit(s)
- Conveyor chain
- Chain conveyor equipment

The chain conveyor plant consists of galvanized chain conveyor sections, built up at the required level (with or without galvanized supports), which are bolted together. Chain profiles are welded into the construction. Chain rails carry special conveyor chains, driven by the necessary number of galvanized driving units (irreversible or reversible) with compact gear. The sections are manufactured of rolled angle profiles and rail profiles, and can be constructed as straight, Y-shaped, V-shaped, or curved sections.

Depending on the chain conveyor plant layout, parts of the plant may consist of roller conveyors. These sections can be constructed as straight or curved sections.

The chain conveyor sections can either be bolted direct onto the floor or mounted with supports of equal or variable height, in order to ensure an adequate working height, and to allow rises and falls of the chain conveyor.

The chain conveyor sections are constructed in standard widths, which are chosen according to the actual cylinder diameter. If cylinders of different diameters are to be transported in the chain conveyor plant, a universal solution can be chosen according to Kosan Crisplant's standard chain conveyor systems.

By combining the types of sections (see drawings "Examples of chain conveyor systems") a high degree of flexibility is obtained, meaning that basically all requirements on handling of LPG cylinders can be met and can be applied to both new and existing filling and/or maintenance plants.

As the chain conveyor sections are manufactured according to Kosan Crisplant standard, it is always possible to replace, modify, or extend existing plants.

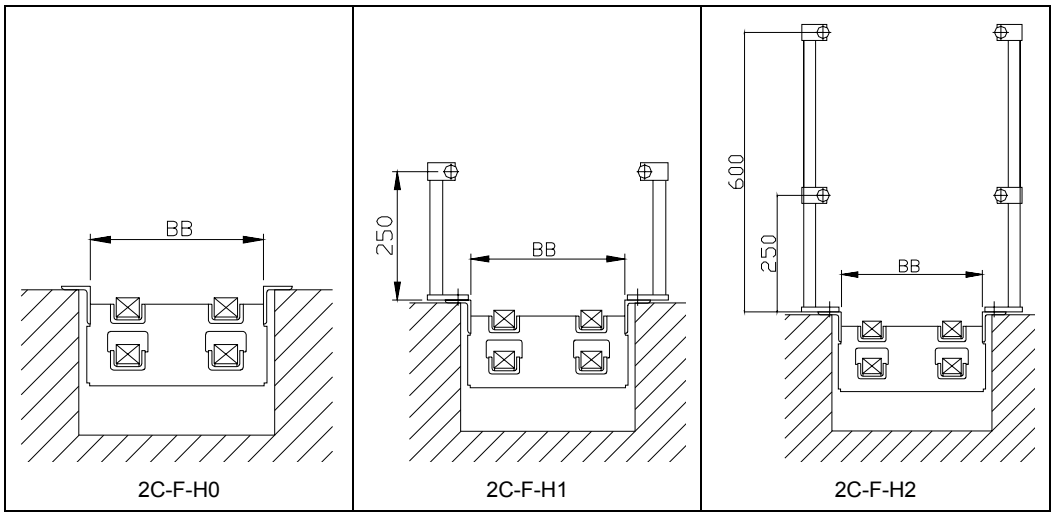
The chain, which is manufactured of steel, is supported by steel slide rails. These can be replaced, thus facilitating maintenance.

The velocity of the conveyor chain is adapted to the actual cylinder capacity, but can be increased or decreased by changing the gear transmission and/or the motor/gear size.

© Kosan Crisplant 2003

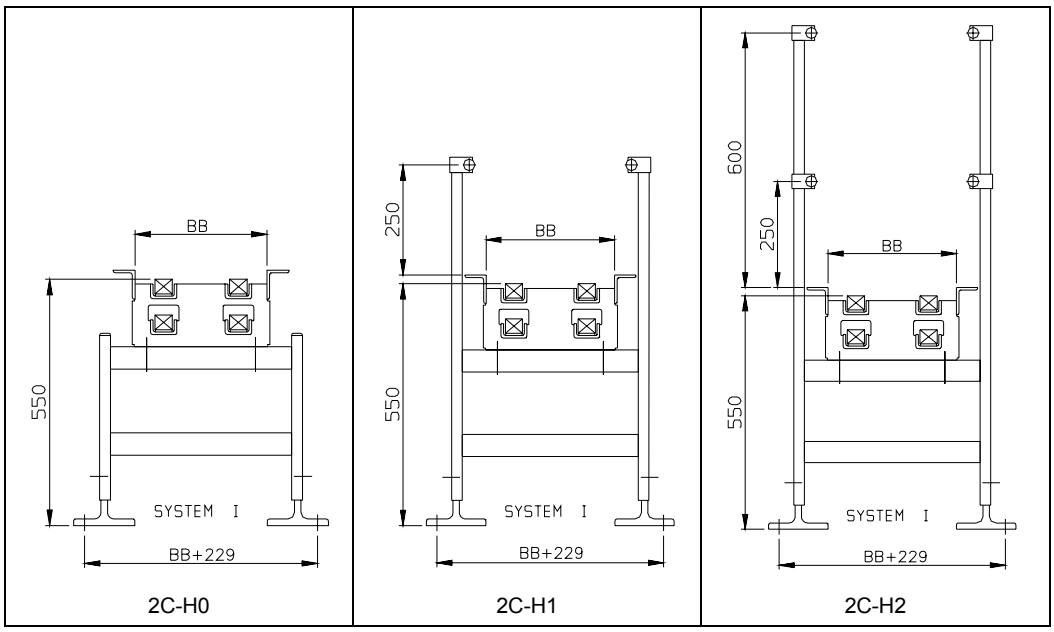
Examples of chain conveyor systems

SYSTEM I (with 2 chains IN floor)



BB = Track width (adjusted according to the actual cylinder bottom ring diameter)

SYSTEM I (with 2 chains ON floor)

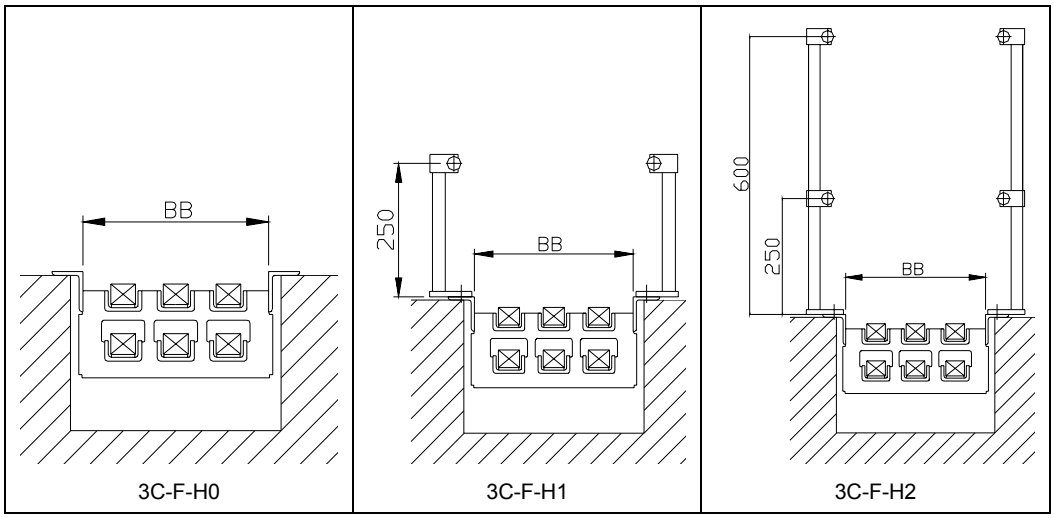


BB = Track width (adjusted according to the actual cylinder bottom ring diameter)

© Kosan Crisplant 2003

Examples of chain conveyor systems

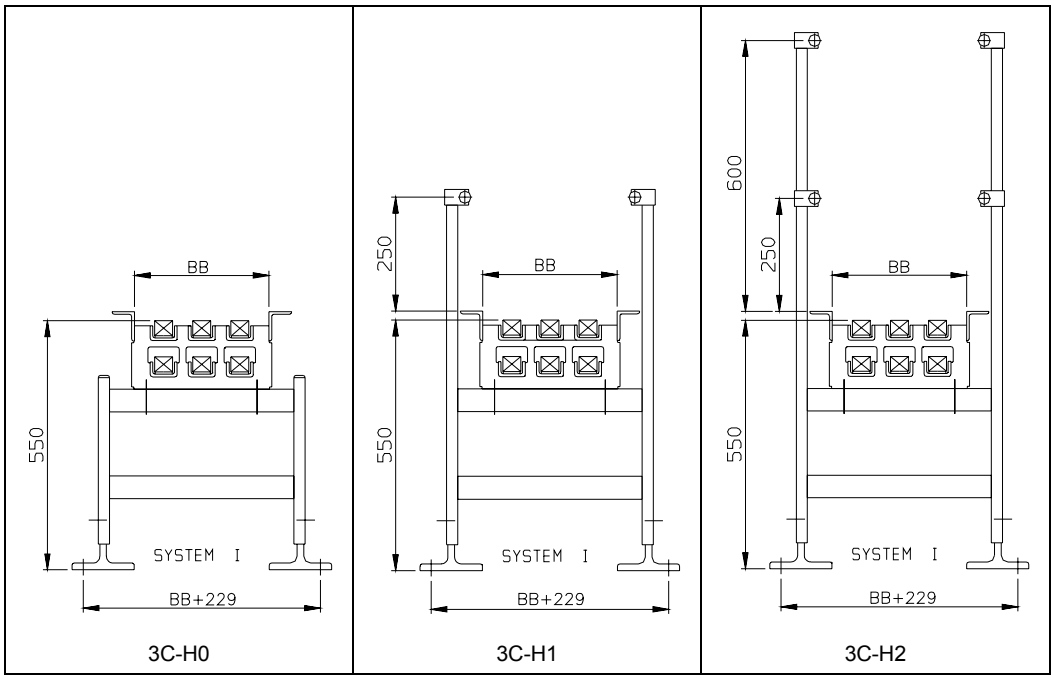
SYSTEM I (with 3 chains IN floor)



3C-F-H0 | 3C-F-H1 | 3C-F-H2

BB = Track width (adjusted according to the actual cylinder bottom ring diameter)

SYSTEM I (with 3 chains ON floor)



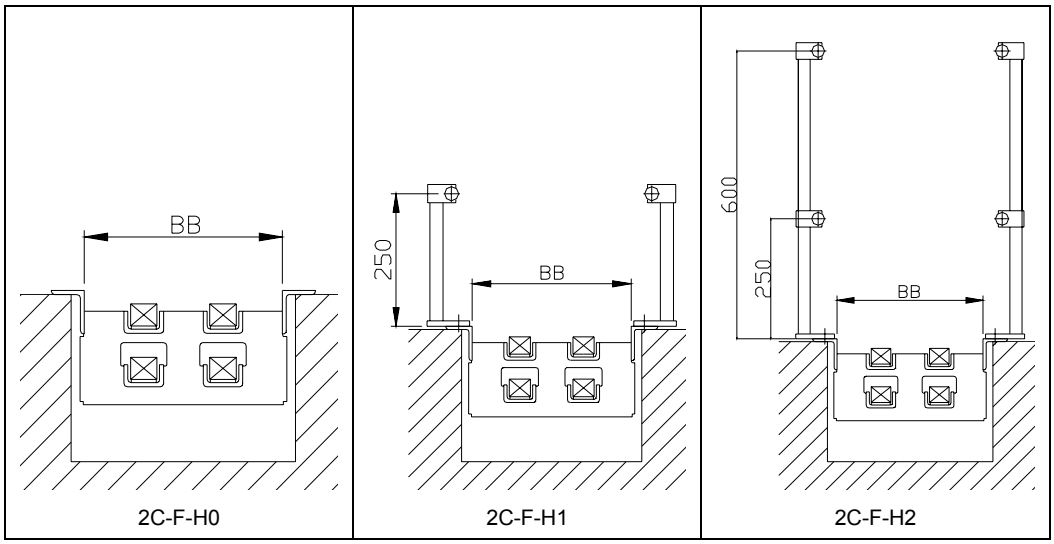
3C-H0 | 3C-H1 | 3C-H2

BB = Track width (adjusted according to the actual cylinder bottom ring diameter)

© Kosan Crisplant 2003

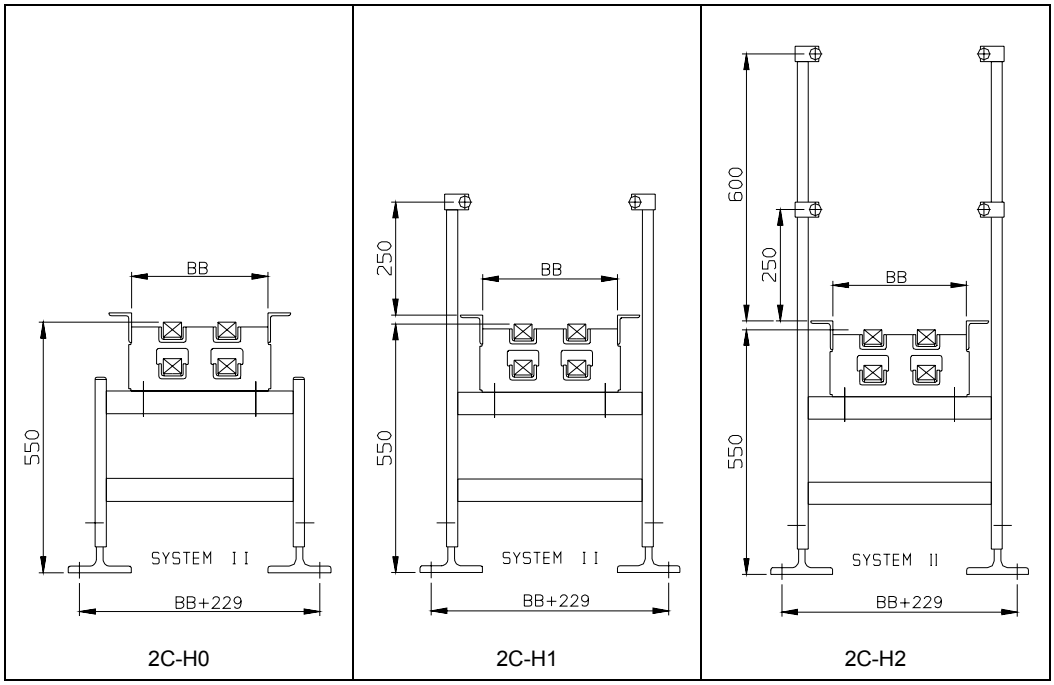
Examples of chain conveyor systems

SYSTEM II (with 2 chains IN floor)



BB = Track width (adjusted according to the actual cylinder bottom ring diameter)

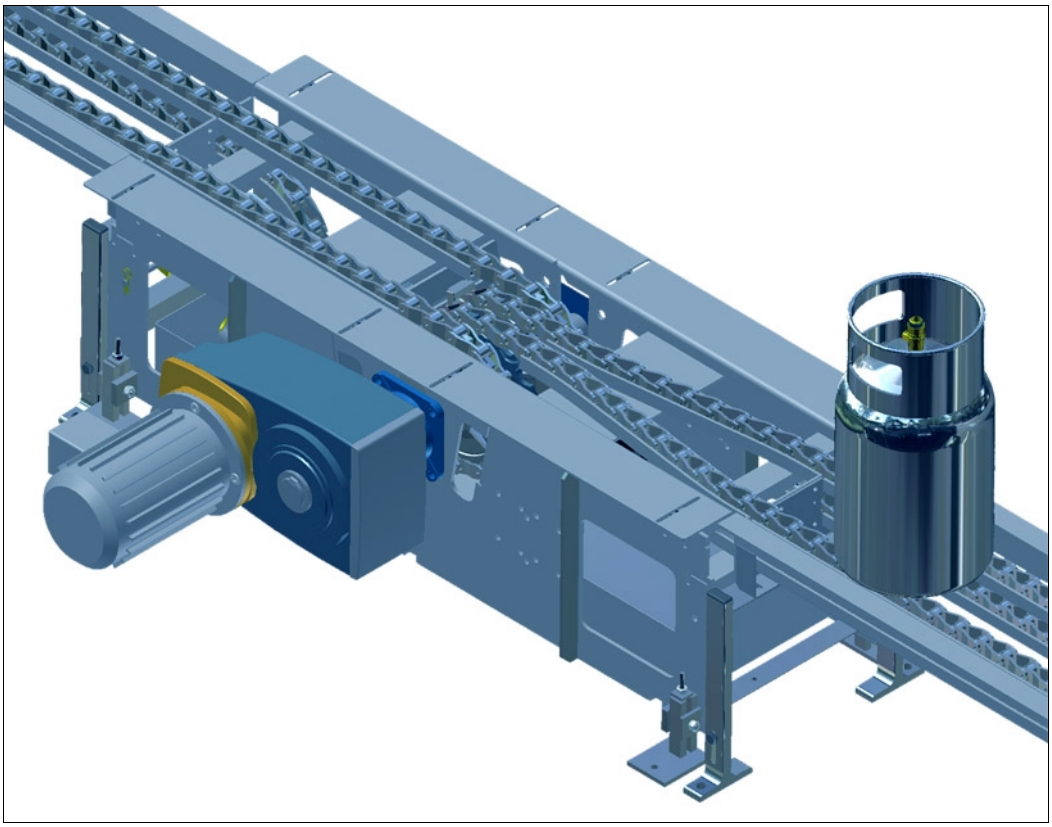
SYSTEM II (with 2 chains ON floor)



BB = Track width (adjusted according to the actual cylinder bottom ring diameter)

© Kosan Crisplant 2003

Shaft gear for driving unit



A chain conveyor plant is equipped with standard galvanized driving units to drive the chain.

The driving unit incorporates special gear motors, belonging under the group of compact gears, which comply with demands for great reliability as well as environmental demands in respect of noise level. By using standard mounting elements for the different gear types and sizes, the number of variations in design can be reduced to a minimum.

Bearings are not built into screwed-on covers and flanges, as they would cause displacements and, consequently, leaks due to wear. The bearings are therefore easy to replace, resulting in a minimum demand for service.

The gears can be reversed by exchanging the air screw and the oil drain screw, and they are intended for mounting on a flanged motor with B5 frame and the number of rpm according to the local frequency (50 or 60 Hz).

The gears are delivered filled up with oil. Oil change is required for every 10,000 hours' normal running.

© Kosan Crisplant 2003

Conveyor chain



The conveyor chain, type 20.000, is specially designed for transport of LPG cylinders in chain conveyor plants. The renowned quality chain from Kosan Crisplant has been supplied to customers in virtually any country in the world that houses a chain conveyor plant for LPG cylinders.

The components of the chain are manufactured of special steel, assuring great wear resistance, low friction, and great tensile and ultimate strength. Furthermore, most of the materials used in the fabrication of the original Kosan Crisplant transport chain are duly certified to ensure the well-known Kosan Crisplant quality.

All processes in the chain component production phase are monitored, which guarantees that each component is 100% perfect before assembly. Defective components are rejected and immediately scrapped.

In addition, random samples of the ready-made chain sections (standard length 5 m) are taken to make absolutely sure that the chain meets the strict Kosan Crisplant quality standard. For each produced batch, spot tests of the chain are made for load on the chain rivets in the form of testing of tensile strength and rivet press-out strength.

© Kosan Crisplant 2003

Chain conveyor equipment

The chain conveyor equipment is designed for guiding and regulating the flow of the cylinders, which are conveyed on the chain conveyor plant.

The equipment can be controlled manually, pneumatically or electronically, and comprises the following range of products/components:

- **Pneumatic activators and photocells**:
  Sends impulses produced by the passage of cylinders to machines/functions located elsewhere in the chain conveyor plant.
- **Pneumatically controlled chain conveyor shifter**:
  Designed for automatic diversion of cylinders into different chain conveyor directions.
- **Manually operated chain conveyor shifter**:
  Designed for manual change of the flow of cylinders on the chain conveyor.
- **Counter system**:
  Designed for either dosing or simply counting a predetermined number of cylinders. A counter system receives signals either from a pneumatic activator or a photocell.
- **Pneumatically controlled or manually operated cylinder stops**:
  Designed for stopping, detaining, or releasing cylinders at intervals.
- **Pneumatically controlled pushing devices**:
  Are located at points where cylinders are sorted out for other operations.
- **Air service unit**:
  Is used where pneumatic operations are carried out. The unit is connected to the supply of compressed air. The unit ensures that the pilot pressure air is filtered, lubricated in case this is prescribed, and the pressure is regulated.
- **Cylinder railings and banisters**:
  Height level according to requirements.
- **Control desk**:
  Equipped with main valves, is connected to chain conveyor equipment, such as cylinder stops, chain conveyor shifters, and pushing devices. The control desk is located at the chain conveyor, centrally in the cylinder flow of the plant.

Scope of supply

The chain conveyor plant is delivered complete according to layout and quotation, with the necessary number of galvanized chain conveyor sections, roller conveyors, and driving units, including conveyor chain to the required extent (in lenghts of 5 m), and the necessary chain conveyor equipment.

Additional equipment

(Only included if specified in quotation)

- Assembly kit for chain conveyor sections (bolts, nuts and washers), including equalizing connections (isolated copper cables)
- Assembly kit for supports (expansion bolts)
- Special tool kit for assembly of 5 m chain lengths (the tools can also be used for disassembly of chain links in case it becomes necessary to shorten the chain
- Standard spare parts set

© Kosan Crisplant 2003

Technical data

Chain conveyor speed	• Always adjusted to the required capacity (normally within the range of 8 to 15 m/min, but the chain conveyor can be adjusted to basically any speed required)
Necessary connections (specifications and quality requirements)	• Compressed air installation: – Working pressure: Min. 0.5 MPa. (5 bar) - Max. 0.8 MPa. (8 bar) – Quality: The quality of the compressed air should be in accordance with ISO 8573.1, class 2-3-2[1)]. • Electrical power installation: – Voltage: 3 x 380/400 VAC – Frequency: 50 or 60 Hz – The installation is performed according to standards EN50014, EN50018, EN50019 and EN50020. • Earth net: – Max. transition resistance between earth wire and earth: ≤ 2.0 Ohm. When assembling the chain conveyor sections the supplied equalizing connections, which have a transition resistance of ≤ 2.0 Ohm, must be mounted at the flanged joints.The chain conveyor plant must be earthed as required according to EN 60 204-1 or API RP 2003. (**Note**! It is important to make sure that the main earthing point on the chain conveyor is connected to the earthing system according to specifications).
Conveyor chain type 20.000	• Allowable chain pull: 1,000 Lbs • Breaking load: 4,000 Lbs • Test load: 3,000 Lbs
Galvanization	• The chain conveyor sections and the driving unit are galvanized with a layer thickness of 80-100 µ

[1)] The compressed air must not contain particles larger than 1 µm. The pressure dew point of the compressed air at a pressure of 0.7 MPa (7 bar) shall be at least 10°C lower than the lowest ambient temperature to which the compressed air system is exposed at the coldest place of the plant during the year (class 3 ~ -20°C). The maximum oil contents in the compressed air must not exceed 0.1 mg/m³.

© Kosan Crisplant 2003